

Step into the World of Code

Syllabus Overview

COURSE INFORMATION:

Course Title:	Robotics and IoT
Target Audience:	Grade 8 and above students, aged 12 and above,
Level:	Beginner to Intermediate
Duration:	30 Classes, 1 Hour Each
Mode of Delivery:	Online or in-person, hands-on learning.
Prerequisites:	No prior experience required

Learning Objectives:

- ♦ Learn how robots work using sensors, actuators, and basic programming.
- ♦ Gain foundational knowledge of robotics systems and IoT integration..
- ♦ Learn C++ basics to control robotic components.
- ♦ Integrate sensor data and control mechanisms through coding.
- ♦ Apply programming to create functional and responsive robotic systems.
- ♦ Use robotics to solve real-world problems in various domains.

UNIT 1 INTRODUCTION TO ROBOTICS & PROGRAMMING

Class 1: Introduction to Robotics

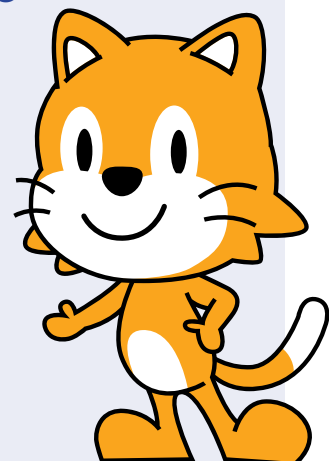
Goal: Understand the basics of robotics and its real-world applications.

Activity: Discuss different types of robots and their uses in various industries.

Class 2: Introduction to C++ for Robotics

Goal: Learn the basics of the C++ programming language.

Activity: Write simple C++ programs to control robot movements.



Class 3: Understanding Sensors and Actuators

Goal: Explore different types of sensors and actuators used in robotics.

Activity: Introduction to basic sensors (e.g., ultrasonic, temperature) and actuators (e.g., motors).



Class 4: Working with Sensors and Data Collection

Goal: Learn how to collect and process data from sensors.

Activity: Program a robot to read data from an ultrasonic sensor and display it.

Class 5: Networking and IoT Architecture

Goal: Introduction to IoT and how robots can connect to networks.

Activity: Set up a basic IoT system where the robot sends data to a web interface.

UNIT 2 ROBOTICS PROJECTS

Class 6: Project 1: Collision Detection

Goal: Program a robot to detect obstacles and avoid collisions.

Activity: Use sensors to program the robot for collision avoidance.

Class 7: Project 1 – Collision Detection (Part 2)

Goal: Program a robot to respond to obstacles.

Activity: Implement the avoidance mechanism using sensor data.

Class 8: Project 2: Soil Moisture Detection

Goal: Program a robot to detect soil moisture for agricultural use.

Activity: Set up soil moisture sensors and program the robot to monitor moisture levels.

Class 9: Project 3: Smart Dustbin

Goal: Design a robot to detect trash and automatically open a dustbin lid.

Activity: Use proximity sensors and motors to automate the dustbin opening process.

Class 10: Project 4 – Smart Home (Part 1)

Goal: Create a basic home automation system.

Activity: Introduce smart home components and plan appliance control logic.

Class 11: Project 4 – Smart Home (Part 2)

Goal: Implement sensor-based home automation features.

Activity: Program appliances to respond to sensor input.

Class 12: Project 4 – Smart Home (Part 3)

Goal: Finalize and test the smart home prototype..

Activity: Integrate all sensors and test complete automation systems..

Class 13: Project 5 – RC Car (Part 1)

Goal: Build the chassis and setup electronic components of an RC car

Activity: Connect motor drivers and wheels to the robot base

Class 14: Project 5 – RC Car (Part 2)

Goal: Program basic movement control for RC Car.

Activity: Develop remote commands to move the car forward, backward, left, right.

Class 15: Project 5 – RC Car (Part 3)

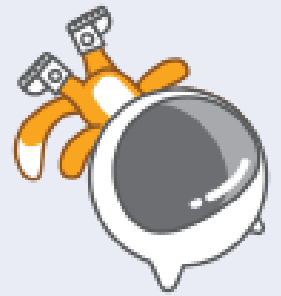
Goal: Final testing and customization

Activity: Add additional features like speed control or lights.

Class 16: Project 6: Smart Parking (Part 1)

Goal: Build a system that helps park a robot in a designated parking spot.

Activity: Program a robot to identify parking spaces and park autonomously.



Class 17: Project 6 – Smart Parking (Part 2)

Goal: Program the robot to park in an identified slot

Activity: Write code for alignment and positioning.

Class 18: Project 7: Automatic Street Light (Part 1)

Goal: Introduce automatic lighting concept using sensors.

Activity: Use an LDR sensor to detect light levels.

Class 19: Project 7 – Automatic Street Light (Part 2)

Goal: Program robot to switch lights based on real-time data.

Activity: Integrate sensor values with relay or LED lights.

Class 20: Project 8 – Line Following Robot (Part 1)

Goal: Understand how IR sensors detect lines.

Activity: Assemble IR sensors and test detection capability.

Class 21: Project 8 – Line Following Robot (Part 2)

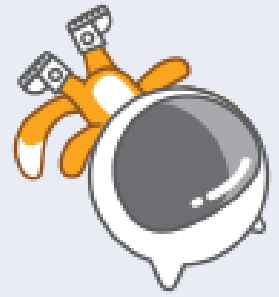
Goal: Implement basic line following behavior.

Activity: Write and test code to follow a black line.

Class 22: Project 8 – Line Following Robot (Part 3)

Goal: Enhance accuracy and speed of line following.

Activity: Tune parameters for sharp turns and speed adjustment.



➡ **Class 23: Project 9 – Gesture-Controlled Robot (Part 1)**

Goal: Introduce gesture detection using sensors (e.g., accelerometers).

Activity: Connect and calibrate gesture module.

➡ **Class 24: Project 9 – Gesture-Controlled Robot (Part 2)**

Goal: Program robot to move based on hand gestures.

Activity: Map gesture data to motion commands.

➡ **Class 25: Project 9 – Gesture-Controlled Robot (Part 3)**

Goal: Optimize gesture controls and finalize the build.

Activity: Test complete gesture system with error handling.

➡ **Class 26: Project 10 – Smart City (Part 1)**

Goal: Plan a smart city prototype layout.

Activity: Brainstorm on city issues like traffic, pollution, waste.

➡ **Class 27: Project 10 – Smart City (Part 2)**

Goal: Develop modules for traffic or environment monitoring.

Activity: Implement sensors to simulate real-time data.

➡ **Class 28: Project 10 – Smart City (Part 3)**

Goal: Integrate modules and enable communication.

Activity: Combine all systems into a unified robot setup

➡ **Class 29: Project 10 – Smart City (Part 4)**

Goal: Final testing and refinement.

Activity: Troubleshoot and enhance performance.

➡ **Class 30: Project 10 – Smart City (Part 5)**

Goal: Showcase and explain the full smart city system.

Activity: Present each feature and demonstrate system working.

MERO
CODING

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